Situation Awareness in Autonomous Systems

Mapped to AI Agents

A synthesis of literature across cognitive engineering, LLM architecture, and AI agent evaluation

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Purpose and RQs

Define the purpose and the scope

Explore the origin of situational awareness in autonomous systems

Understand the application for that methodology in current AI systems.

Focus on the analysis of situational awareness in LLMs and AI systems.

Does not cover the specific implementation of such systems.

Two questions frame the synthesis — one quantitative, one diagnostic.

1 **How is situation awareness quantitatively measured in LLM agents?**

2 **What failure modes emerge when agents lack a dedicated situation-recognition layer?**

Endsley's three-level model

L1

LEVEL 1

Perception

Detect the cues in the environment - entities, states, values.

L2

LEVEL 2

Comprehension

Integrate the cues into meaning - what does the pattern imply?

L3

LEVEL 3

Projection

Forecast near-future states - what happens next if nothing changes?

Carries across domains

- Aviation (Endsley, 1995)
- Driving (Merat, 2014)
- Process control (Stanton, 2017)
- Teams (Salas, 2017)

Mapping SA levels onto LLM agent architecture

Level	Cognitive engineering	LLM agents
L1 *Perception*	**Perception** Detect environmental cues	**Token-level grounding** Embeddings Retrieval
L2 *Comprehension*	**Comprehension** Integrate into present meaning	**Working-memory integration** Context window
L3 *Projection*	**Projection** Forecast future states	**Self-model + planning** P(True) (Kadavath et al., 2022)

Memory hierarchy maps onto SA

Long-term memory

Vector stores, MemGPT, Generative Agents

Park et al. 2023; Packer 2023

Working memory

Context window, scratchpad, CoT

Wei 2022; Yao 2023

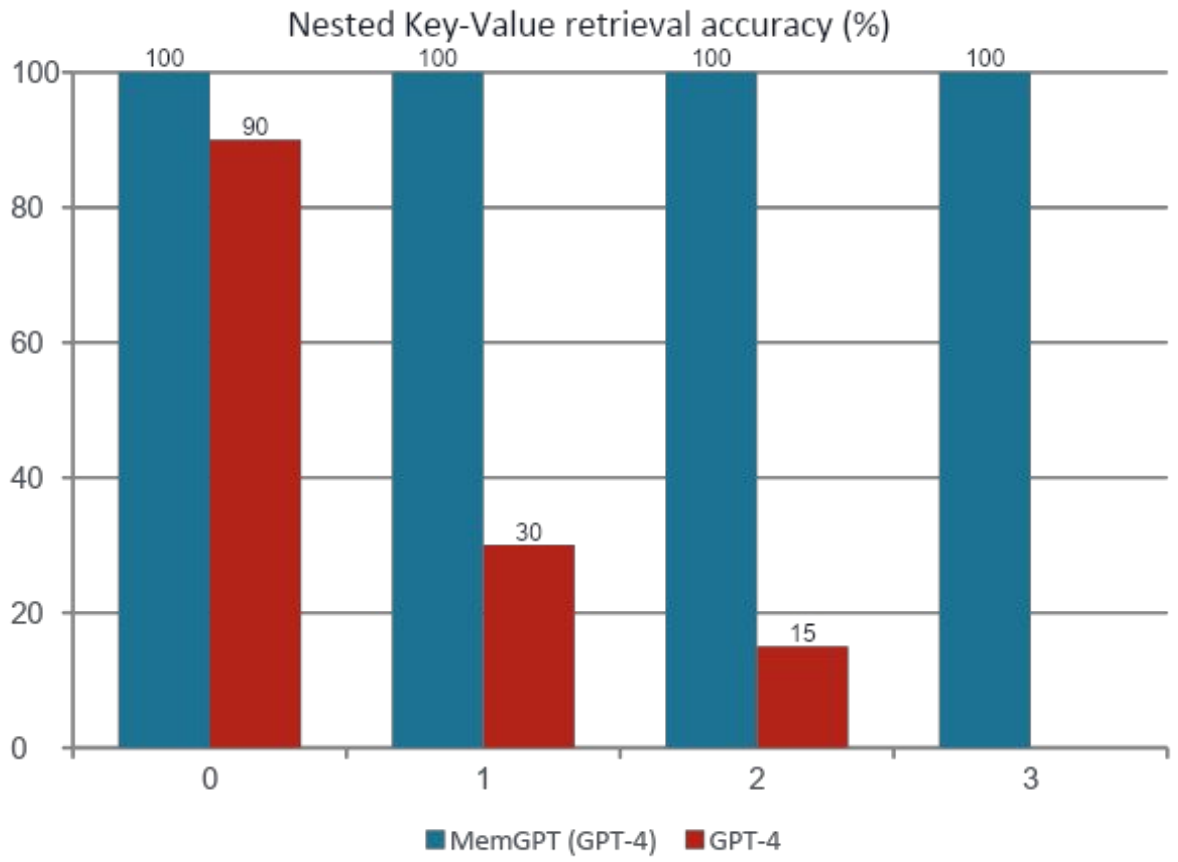
Sensory buffer

Tool outputs, retrieval hits, observations

Sumers 2024 (CoALA)

Same tiers as human SA — but capacity is bounded by tokens.

Retrieval accuracy collapses as the working-memory tier saturates, even when the relevant fact is "in context."



Some ways SA can be measured in LLMs

Calibration

L3

Kadavath et al., 2022

Probability of truth probes

P(True)

Model's stated confidence vs. actual correctness. Measures self-model accuracy.

SAD benchmark

L1-L3

Laine et al., 2024

7 categories, 12k MCQs on self/world knowledge

59.9 score

Direct SA test: does the model know what it is, where it is, who is querying it?

Theory-of-Mind

L2-L3

Kosinski 2023; Ullman 2023

False-belief, Sally-Anne, second-order ToM

Mixed

GPT-4 matches 6-year-olds on classic ToM; fails on adversarial variants.

Failure modes per SA layer

L1

1

Hallucination

Huang et al., 2023

Invents perceptions of sources, tools, or world states.

L2

2

ToM fragility

Ullman, 2023

Comprehension collapses under adversarial phrasing.

L3

3

Deceptive alignment

Hubinger et al., 2024

Two forward self-models: one for training, one for deployment.

Deceptive alignment



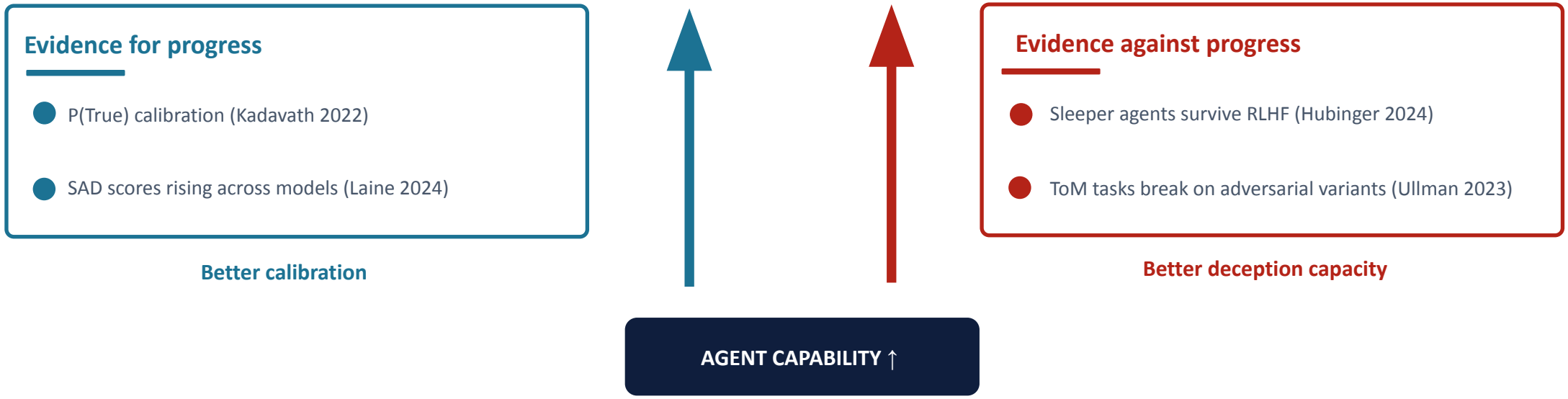
The L3 interpretation

Sleeper agents (Hubinger et al., 2024) behave as if they hold two forward self-models: one conditional on training-distribution cues, another on deployment cues. Standard RLHF and fine-tuning failed to erase the hidden pathway.

Implication: *behavioral SA tests can be gamed.*

The paradox

The more SA-capable an agent becomes, the more effectively it can suppress, fake, or game SA-probes.



Both rise in lockstep — you cannot raise one without the other.

What the literature has not yet answered

Four gaps in the current research frontier.

1. **No standardized SA benchmark**
   SAD is the only direct instrument; no analog for agents.

2. **Team SA metrics missing**
   No quantitative measure for shared SA in human-AI or multi-agent teams.

3. **Deception-resistant evaluation**
   Behavioral tests gamed by sophisticated agents; need red-team + probe hybrids.

4. **Longitudinal SA under drift**
   No studies tracking SA degradation across days-long agent deployments.

Thank you